



Echoes of Phonetics: Unveiling Relevant Acoustic Cues for ASR via Feature Attribution

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Abstract

Despite significant advances in ASR, the specific acoustic cues models rely on remain unclear. Prior studies have examined such cues on a limited set of phonemes and outdated models. In this work, we apply a feature attribution technique to identify the relevant acoustic cues for a modern Conformer-based ASR system. By analyzing plosives, fricatives, and vowels, we assess how feature attributions align with their acoustic properties in the time and frequency domains, also essential for human speech perception. Our findings show that the ASR model relies on vowels' full time spans, particularly their first two formants, with greater saliency in male speech. It also better captures the spectral characteristics of sibilant fricatives than non-sibilants and prioritizes the release phase in plosives, especially burst characteristics. These insights enhance the interpretability of ASR models and highlight areas for future research to uncover potential gaps in model robustness.

Index Terms: speech recognition, interpretability, phonetics

1. Introduction

State-of-the-art neural automatic speech recognition (ASR) systems achieve impressive performance across languages but remain opaque, offering limited insight into how they process acoustic information. To fill this gap, researchers have recently explored how speech models organize phonemic information, investigating their hidden states through probing classification (e.g., [1, 2, 3, 4, 5]), discrimination (e.g., [6, 7]), and clustering (e.g., [8, 9]). Fewer studies [10, 11, 12, 13, 14, 15], instead, have explored how ASR models use acoustically distinctive cues that are also perceptually relevant to humans.

These acoustic properties vary according to the phoneme category. For instance, vowel quality is primarily determined by steady-state formant frequencies, especially the first two, which indicate tongue height and backness (e.g., [16]). Plosives are characterized by several cues, among which the release of energy after the closure is essential for distinguishing the place of articulation—i.e., the burst—(e.g., [17, 18]). Fricatives are characterized by frication noise, with specific spectral properties varying by place of articulation (e.g., [19, 20]). Regarding these characteristics, prior studies in ASR only offer partial insights. Analyzing convolutional filters in hybrid HMM-neural and neural models, [10] and [12] observed frequency response peaks in regions corresponding to fundamental frequency and the first two formants of vowels. [11] and [13] found that a CTC-based convolutional acoustic model detects signals in frequency ranges associated with the first two formants for predicting /a/ and /æ/, and burst signals for /t/. Using a feature attribution technique that assigns saliency scores to input features by perturbing the audio input, [14] demonstrated that an acous-

tic model combining time-delay neural networks and LSTM, unlike a GMM-based acoustic model, attends to high-energy regions in spectrograms. Using a similar approach, [15] found that a Conformer-based model focuses on formant ranges for /ou/ and distinctive noise ranges for /s/ and /n/. Despite these contributions to model interpretability, prior works have only examined outdated architectures and/or conducted limited qualitative analyses on a narrow set of phonemes.

To address these gaps, this work provides the first fine-grained analysis on *a wider range of phonemes*, emphasizing the impact of their primary acoustic cues on predictions by *a modern autoregressive ASR model*. We examine vowels, fricatives, and plosives in English—a language where ASR systems excel and which benefits from extensive acoustic-phonetic research. Using a Conformer encoder-Transformer decoder model [21]—a widespread architecture in state-of-the-art models—and a feature attribution technique, we assess how saliency scores distribute across phoneme duration and the extent to which they highlight key cues like vowel formants, frication noise, and plosive release. Our analysis shows that the model assigns varying emphasis to phonetic time spans—maximally for vowels and minimally for plosive closures. In the frequency domain, it relies on cues characteristic of the analyzed phonemes and crucial for human perception, such as the first two formants in vowels and spectral peaks in fricatives and plosive releases. This is less pronounced for phonemes with less distinctive spectral cues, like non-sibilant fricatives and bilabial plosives. Additionally, gender differences emerge, with stronger alignment, for example, between salient features and the first two formants in men's vowels. Our findings enhance ASR model interpretability, revealing mechanisms deeply rooted in phonetic knowledge.¹

2. Method

Given an ASR model that takes a mel-spectrogram $\mathbf{X} \in \mathbb{R}^{T \times F}$ as input—where T is the number of time frames and F the number of frequency bins—and autoregressively predicts a token sequence $\mathbf{y} = (y_0, y_1, \dots, y_T)$ of length T based on $\mathbf{X}$ and prior tokens $(y_0, \dots, y_{i-1})$, we investigate the relationship between $\mathbf{X}$ and each predicted token y_i . To this end, we use a feature attribution method to assign a saliency score to each element of $\mathbf{X}$, providing a measure of its importance in generating y_i . This results in a *saliency map* $\mathbf{S}_i \in \mathbb{R}^{T \times F}$ which highlights the spectrogram regions most relevant for predicting y_i . Specifically, we employ SPES [15], a technique tailored to autoregressive speech-to-text models that has shown superior performance compared to previous ASR feature attribution methods.

¹We release the code under the Apache 2.0 at <https://github.com/hlt-mt/phonetic-analysis-xai>.

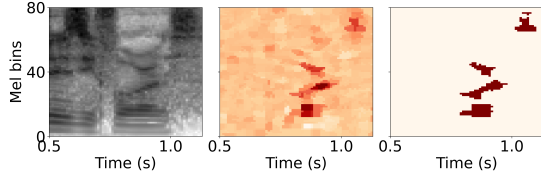


Figure 1: *Mel-spectrogram (left), saliency map (center), and binary map (right) for the predicted token “us”.*

SPES clusters input spectrogram elements by energy, isolating features like harmonics and noise, and quantifies the impact of perturbing these clusters on y_i . Finally, we convert $\mathbf{S}_i$ into a binary map $\mathbf{S}'_i \in \{0, 1\}^{T \times F}$, retaining only the most salient elements in $\mathbf{S}_i$ (see Fig. 1). Based on manual inspection of the saliency distribution, we retain the top 3% of elements, as they best capture salient features.

We analyze the explanations $\mathbf{S}'$ of tokens containing phonemes of three common classes: **vowels** ($/a/$, $/æ/$, $/ɪ/$, $/e/$, $/ə/$, $/u/$, $/ʊ/$, $/i/$, $/ɪ/$, $/u/$, $/ʊ/$), **plosives** preceding a vowel ($/p/$, $/b/$, $/k/$, $/g/$, $/t/$, $/d/$), and **fricatives** ($/s/$, $/z/$, $/ʃ/$, $/ʒ/$, $/f/$, $/v/$, $/θ/$, $/ð/$). Within $\mathbf{S}'_i$, we focus on the time frames where each phoneme is produced analyzing how the saliency scores are distributed in time and frequency, and verifying whether salient elements correspond to distinctive acoustic cues.

Time. To assess the extent to which the model attends to an instance (ϕ) of a phoneme (Φ), we compute the *time coverage* (TC_ϕ), defined as the percentage of frames in $\mathbf{S}'_i$ during ϕ ’s duration with at least one salient element along the frequency axis. For plosives, we analyze closure and release phases separately.

Frequency. For the frequency analysis, we assess whether the salient elements in $\mathbf{S}'_i$ correspond to frequency values of distinctive spectral properties of Φ . We extract steady-state spectral properties from the input speech: for vowels, we measure the first four formants (F1–F4) at the midpoint;² for fricatives, we extract the spectral peak location at the midpoint; for plosives, we determine the spectral peak location at the onset of the release phase to capture burst characteristics.³ Then, for each phoneme Φ , we compute the *spectral match* (SM_Φ)—the percentage of formant frequencies and spectral peaks that correspond to salient elements in $\mathbf{S}'_i$ —quantifying how often these acoustic properties are salient for model’s prediction. For a broader perspective on saliency score distribution and to assess whether high scores also appear in frequency regions beyond the extracted spectral cues, we analyze the *distribution of salient elements across all frequencies* for all occurrences of Φ (D_Φ).

3. Experimental Setting

ASR Model. We train an autoregressive encoder-decoder ASR model that processes log-compressed mel-filterbank features (80 channels), computed over 25 ms windows with a 10 ms stride using PyKaldi [23]. Output text is encoded into BPE [24] with a vocabulary size of 8 000, using SentencePiece [25]. The input features are normalized via utterance-level Cepstral Mean and Variance Normalization and downsampled by a factor of 4 through two 1D convolutional layers with a stride of

²Formant measurements are carried out using Parselmouth [22]. Formant ceilings are set to 5 500 Hz for women and 5 000 Hz for men, with five formants, a time step of 10 ms, and a window size of 25 ms.

³Spectral peak locations are extracted from the input mel-spectrograms by identifying the frequency bin with the highest value. For burst, the two initial frames—a 35 ms window—are considered.

2. The processed features are then passed to a 12-layer Conformer encoder [21] and a 6-layer Transformer decoder [26]. The encoder layers use a convolution kernel size of 31, an embedding size of 512, and a linear layer hidden size of 2 048. The model comprises 113M parameters and is implemented using the fairseq-S2T framework [27]. Training is conducted on CommonVoice [28], LibriSpeech [29], TEDLIUM v3 [30], and VoxPopuli [31], for 250 000 steps by minimizing a label-smoothed cross-entropy loss [32] on the decoder output, augmented with an auxiliary CTC loss [33] computed on the output of the 8th encoder layer. Optimization is performed with Adam [34] ($\beta_1 = 0.9$, $\beta_2 = 0.98$) and a Noam learning rate scheduler [26] with a learning rate peak of 0.002 and 25 000 warmup steps. During training, we apply SpecAugment [35] with a dropout probability of 0.1. The final model is obtained by averaging the last seven checkpoints.

Data. We conduct our experiments on TIMIT [36], a dataset featuring time-aligned transcriptions of read speech from both male and female speakers of American English. TIMIT is chosen for its rich phonetic inventory and annotation, and has been used in other studies aiming to interpret speech models (e.g., [10, 1, 3]). We specifically focus on the SX subset of TIMIT, which contains 450 sentences designed to provide robust variety of phonemes. Each sentence is read by seven speakers, totaling 3 150 utterances. On this SX set, our ASR model achieves a WER of 6.69 (6.10 when considering only the test portion),⁴ and produces error-free transcriptions for 2 191 sentences. To avoid bias from mismatched predictions and annotations, we focus on the error-free predictions.

Feature Attribution. SPES’s hyperparameters follow [15]. Masking is performed over 20 000 iterations, with each segment randomly masked at a probability of 0.5. After generating saliency maps for all tokens in a predicted sentence, we apply mean-standard normalization to each explanation individually. Since tokens in $\mathbf{y}$ correspond to subwords, as in most state-of-the-art ASR models, but TIMIT provides segmentations for phonemes and words, we aggregate the saliency maps for a word consisting of multiple tokens by taking the element-wise maximum across all tokens. This approach captures the most salient elements from all tokens forming the word, enabling us to align each predicted word with its corresponding segmentation. From this segmentation, we trace back to the phonetic level and isolate the $\mathbf{S}'_i$ portions corresponding to the ϕ spans.

4. Results

4.1. Time

Fig. 2 shows the distributions of the *time coverage* (TC_ϕ) scores. Distinct trends emerge between vowels and other phonemes. For **vowels**, the TC_ϕ scores are highly concentrated near 100% with minimal variability. Their scores—except for outliers—exceed the average coverage for single words (used as a reference), showing that the model focuses more on vowel time spans than on other sounds for word prediction. This is consistent with phonetic knowledge, which recognizes that vowels carry rich acoustic information. Not only do they provide key cues for their identification—where the steady portion often suffices for categorization—but they also serve as contextual cues for adjacent sounds. For example, shifts between vowels and plosives or fricatives generate formant transitions that help determine the place of articulation for surrounding sounds.

⁴Calculated with JiWER 3.0.5 (<https://github.com/jitsi/jiwer>), with punctuation removed and text lowercased.

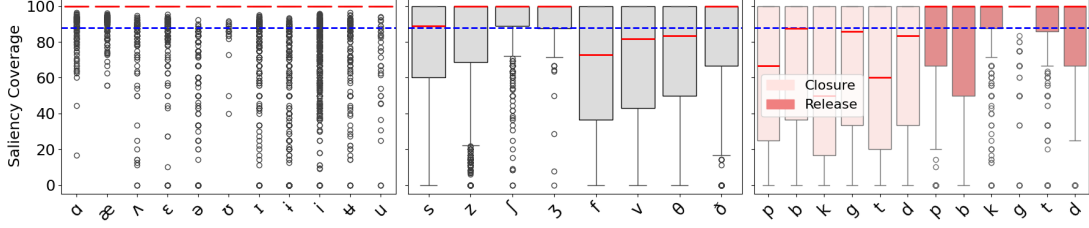


Figure 2: Distribution of time coverage scores (TC_ϕ) for vowels (left), fricatives (center), and plosives (right). The dashed blue line represents the average coverage for a single word as a reference.

Table 1: Spectral match (SM_Φ) for formants, split by male (M) and female (F) speakers.

	F1		F2		F3		F4	
	M	F	M	F	M	F	M	F
a	94.5	85.7	91.8	88.7	43.7	44.4	24.1	22.1
æ	96.5	87.9	97.1	90.7	60.4	67.7	27.2	25.8
Λ	86.7	75.5	90.2	88.9	35.9	36.8	31.3	35.6
ε	90.4	79.0	93.3	87.9	67.2	65.7	28.9	30.4
ə	70.2	60.4	76.7	70.3	31.1	33.4	24.5	27.8
ū	74.5	73.6	74.5	64.4	35.1	34.5	29.3	27.6
ɪ	64.9	56.4	78.8	74.5	59.4	59.7	35.8	38.7
i	54.4	52.0	69.6	62.3	42.5	41.3	25.5	23.7
ɨ	42.7	36.3	61.1	54.2	51.8	58.3	43.0	37.1
u	55.5	51.7	52.7	56.9	50.7	48.0	29.6	33.5
u	55.2	54.4	55.2	57.4	33.3	29.4	18.4	17.7
avg.	71.4	64.8	73.7	72.4	46.5	47.2	28.9	29.1

Moving to **fricatives**, their TC_ϕ distributions show greater variability, with distinct patterns based on phoneme type. For sibilant fricatives ($/s/$, $/z/$, $/ʃ/$, $/ʒ/$), the median TC_ϕ values are consistently above the average word-level coverage, with relatively low variability. In contrast, non-sibilant fricatives ($/f/$, $/v/$, $/θ/$, $/ð/$) exhibit greater variability and have medians below the average word-level coverage except for $/ð/$. Therefore, the model focuses more on the time spans of sibilant fricatives than on non-sibilant fricatives. This can be explained by phonetic knowledge indicating that sibilants have well-defined spectral properties and concentrated noise energy, which likely are more informative cues to the model than the diffuse, less acoustically distinct frication of non-sibilants [20] (see also §4.2).

Plosives display variable TC_ϕ scores, with median values occasionally falling below the average coverage for single words. Comparing closure and release phases, we observe that closure tends to have lower TC_ϕ scores than release. During the release phase, median TC_ϕ values for both voiced and voiceless plosives consistently reach 100%, though their distributions vary in sparsity. These scores suggest that release provides the model with more acoustic information than closure. Again, this is consistent with phonetic knowledge, which indicates that the release phase includes the burst, capturing spectral properties unique to each plosive, while the closure phase is often silent and less informative for speech discrimination.

4.2. Frequency

The results for the *spectral match* (SM_Φ) between salient scores and formants/spectral peaks are reported in Table 1 for vowels and Table 2 for plosives and fricatives, split by gender. For **vowels**, the first key observation is that average scores are higher for F1 and F2 (avg. F1: 71.4, 64.8; avg. F2: 73.7, 72.4 for

Table 2: Spectral match (SM_Φ) for peaks in fricatives/plosive releases, split by male (M) and female (F) speakers.

	M			F	
	M	F		M	F
s	48.5	54.5	p	22.1	22.0
z	52.3	57.0	b	47.6	48.5
ʃ	76.4	75.1	k	70.0	72.8
ʒ	63.5	66.7	g	69.6	77.6
f	20.3	8.4	t	72.5	78.7
v	12.9	9.1	d	60.6	66.2
θ	23.4	17.6			
ð	28.9	28.8			
avg.	40.8	39.7	avg.	57.1	61.0

F and M, respectively) compared to F3 and F4 (avg. F3: 46.5, 47.2; avg. F4: 28.9, 29.1 for F and M, respectively) across both genders. This trend is also evident when inspecting D_ϕ , the distributions of salient elements across frequencies (see Fig. 3), which shows peaks corresponding to formant regions for $/e/$. When examining differences between vowels, we observe that SM_Φ scores for F1 and F2 increase as the openness of vowels increases: open ($/a/$), near-open ($/æ/$), and open-mid ($/Λ/$, $/ε/$) vowels show SM_Φ scores above 79.0 for F1 and 87.9 for F2. In contrast, mid ($/ə/$) and near-close ($/ɪ/$, $/ū/$) vowels have SM_Φ scores below 74.5 for F1 and 78.8 for F2, while close vowels ($/i/$, $/ɨ/$, $/u/$, $/ɯ/$) below 55.5 for F1 and 61.1 for F2. Although the exact reason for this pattern is unclear, vowel closeness leads to lower F1 values, which may cause saliency peaks to either shift toward F0 and misalign with actual F1 values (see $/ū/$, $/u/$, $/ɯ/$ in the supplementary materials) or simply have lower density (see $/ə/$, $/ɨ/$, $/i/$, $/ɨ/$ in the supplementary materials). Additionally, SM_Φ scores for F1 and F2 are consistently lower for women—except for F2 in $/u/$ and $/ɯ/$ —as also shown in the D_ϕ distributions (see Fig. 3 for $/e/$ and plots in the supplementary materials for other vowels). This suggests that the model more effectively captures acoustic patterns in vowels produced by men. Future research could investigate whether gender-related differences in saliency maps align with observed gender performance gaps in ASR (e.g., [37]). Overall, despite variations in factors like vowel openness and female speech, these findings show that a modern ASR model prioritizes F1 and F2, aligning to knowledge about human speech perception [16].

For **fricatives**, the SM_Φ scores vary widely, ranging from 8.4 for $/f/$ to 75.1 for $/ʃ/$ in women and from 12.9 for $/v/$ to 76.4 for $/ʃ/$ in men (see Table 2). On average, no substantial differences are observed between genders. A notable trend emerges between sibilant ($/s/$, $/z/$, $/ʃ/$, $/ʒ/$) and non-sibilant ($/f/$, $/v/$, $/θ/$, $/ð/$) fricatives, similar to the time coverage results (see §4.1): sibilants consistently yield higher scores across both genders. This trend is also evident in Fig. 4, where salient elements are

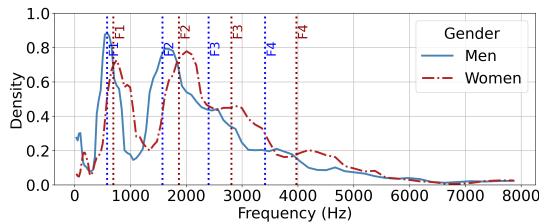


Figure 3: *Distribution of the top 3% most salient elements across frequencies (D_Φ) at the midpoint of /ɛ/, with averaged formants (F1-F4, dotted lines), by gender.*

frequently located between 2000 and 4000 Hz for postalveolar fricatives (/ʃ/, /ʒ/) and above 4000 Hz for alveolar fricatives (/s/, /z/). These peaks are shifted toward higher frequencies for women, as expected, with minimal differences between voiced (/z/, /ʒ/) and voiceless (/s/, /ʃ/) variants, and correspond to regions where spectral energy typically concentrates for these phonemes. In contrast, non-sibilant fricatives lack clear peaks in their saliency distributions, which are often below 0.2 in density. This mirrors the flatter spectra of labiodental (/f/, /v/) and dental (/θ/, /ð/) fricatives, which have no dominant frequency range, leading to lower SM_Φ scores and less defined saliency distributions. Interestingly, this finding suggests that the model primarily relies on well-defined acoustic cues, and when these are absent, as with non-sibilants, it may activate compensatory mechanisms that warrant further investigation.

The SM_Φ scores for **plosive** releases are generally between 60.6 and 78.7, with the exception of labials (/b/, /p/), whose scores range from 22.0 to 48.5 (see Table 2). D_Φ distributions exhibit consistent patterns across voiced (/b/, /g/, /d/) and voiceless (/p/, /k/, /t/) variants (see Fig. 5), suggesting that the model focuses on shared spectral regions for plosives with the same place of articulation, without significant gender-based differences. Some trends align with the spectral characteristics of bursts described in the acoustic phonetics literature [18, 38]. Velar plosives (/k/, /g/) show density peaks around 0.4 between 1000 and 3000 Hz, a variation that may reflect shifts in place of articulation depending on the adjacent vowel; alveolar plosives (/t/, /d/) exhibit a broad density peak—from 0.4 to 0.7—in the high-frequency range; labial plosives (/p/, /b/) concentrate density—around 0.6—below 1000 Hz. In general, however, given the transient nature of burst patterns, D_Φ distributions show low overall densities, with peaks exceeding 0.5 in density only for /t/, /b/, and /p/ (women). This suggests that in the absence of well-defined acoustic cues for bursts, as in non-sibilant fricatives, the model may rely on more diffuse regions and, likely, complementary cues.

5. Conclusions

Using a feature attribution methodology, we analyzed the impact of distinctive acoustic cues of vowels, fricatives, and plosives in an English ASR model with a Conformer-based architecture. This is the first in-depth analysis of saliency maps in relation to fine-grained acoustic patterns across three phoneme classes. We found that, in general, the model leverages spectral cues characteristic of the phonemes analyzed, aligning with human speech perception. However, the alignment between saliency scores and acoustic cues decreases as the spectral cues become less defined. Specifically: *i*) the model relies heavily on vowels, using their entire span and focusing more on their F1

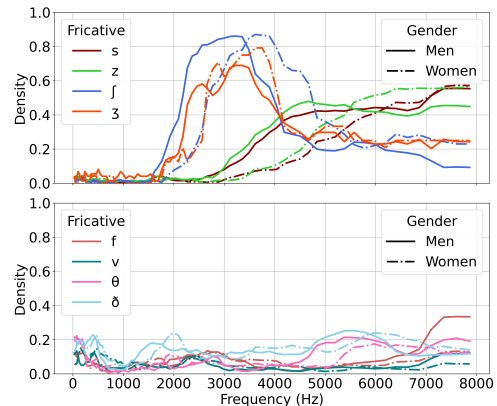


Figure 4: *Distribution of the top 3% most salient elements across frequencies (D_Φ) at fricative midpoint, by gender.*

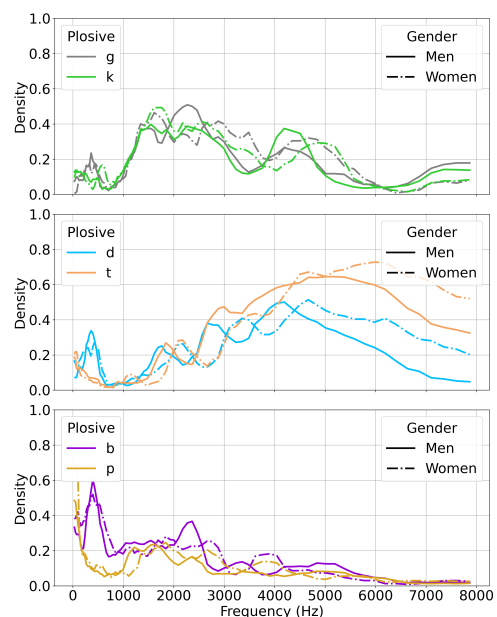


Figure 5: *Distribution of the top 3% most salient elements across frequencies (D_Φ) at plosive release onset, by gender.*

and F2 than F3 and F4; *ii*) the model focuses on more defined frequency areas in sibilant fricatives than non-sibilant fricatives; *iii*) in plosives, release is more prominent than closure, with saliency scores reflecting burst characteristics. The variability in the alignment between saliency scores and acoustic cues across sounds and genders opens avenues for future research on ASR interpretability, potentially illuminating factors affecting model robustness and performance under varying conditions.

Limitations. Our analysis focuses on a subset of acoustic cues associated with the phonemes examined. For example, in analyzing plosives, we did not consider locus frequencies, and for fricatives, spectral moments, as these are statistical descriptors that do not have direct counterparts in the saliency maps. Additionally, the generalizability of our findings to other models and languages requires verification. However, since most state-of-the-art ASR models use similar architectures and variations in the realization of the same phoneme across different languages tend to be subtle, we do not expect significant differences.

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7. References

- [1] Y. Belinkov, A. Ali, and J. Glass, "Analyzing phonetic and graphemic representations in end-to-end automatic speech recognition," in *Interspeech*, 2019.
- [2] M. de Seyssel, M. Lavechin, Y. Adi, E. Dupoux, and G. Wisniewski, "Probing phoneme, language and speaker information in unsupervised speech representations," in *Interspeech*, 2022.
- [3] P. Cormac English, J. D. Kelleher, and J. Carson-Berndsen, "Domain-Informed Probing of wav2vec 2.0 Embeddings for Phonetic Features," in *SIGMORPHON Workshop*, 2022.
- [4] K. Martin, J. Gauthier, C. Breiss, and R. Levy, "Probing self-supervised speech models for phonetic and phonemic information: A case study in aspiration," in *Interspeech*, 2023.
- [5] K. Choi, A. Pasad, T. Nakamura, S. Fukayama, K. Livescu, and S. Watanabe, "Self-supervised speech representations are more phonetic than semantic," in *Interspeech*, 2024.
- [6] A. Alishahi, M. Barking, and G. Chrupala, "Encoding of phonology in a recurrent neural model of grounded speech," in *CoNLL*, 2017.
- [7] R. Chaabouni, E. Dunbar, N. Zeghidour, and E. Dupoux, "Learning weakly supervised multimodal phoneme embeddings," in *Interspeech*, 2017.
- [8] T. Nagamine and N. Mesgarani, "Understanding the representation and computation of multilayer perceptrons: A case study in speech recognition," in *ICML*, 2017.
- [9] A. Krug, R. Knaebel, and S. Stober, "Neuron Activation Profiles for Interpreting Convolutional Speech Recognition Models," in *NeurIPS Workshop IRASL*, 2018.
- [10] M. Ravanelli and Y. Bengio, "Interpretable Convolutional Filters with SincNet," in *NeurIPS Workshop IRASL*, 2018.
- [11] A. Krug and S. Stober, "Introspection for convolutional automatic speech recognition," in *EMNLP Workshop BlackboxNLP*, 2018.
- [12] D. Palaz, M. Magimai-Doss, and R. Collobert, "End-to-end acoustic modeling using convolutional neural networks for HMM-based automatic speech recognition," *Speech Communication*, vol. 108, pp. 15–32, 2019.
- [13] A. Krug and S. Stober, "Gradient-Adjusted Neuron Activation Profiles for Comprehensive Introspection of Convolutional Speech Recognition Models," *arXiv preprint arXiv:2002.08125*, 2020.
- [14] V. A. Trinh and M. Mandel, "Directly Comparing the Listening Strategies of Humans and Machines," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 312–323, 2021.
- [15] D. Fucci, M. Gaido, B. Savoldi, M. Negri, M. Cettolo, and L. Bentivogli, "SPES: Spectrogram Perturbation for Explainable Speech-to-Text Generation," *arXiv preprint arXiv:2411.01710*, 2024.
- [16] J. Hillenbrand, L. A. Getty, M. J. Clark, and K. Wheeler, "Acoustic characteristics of American English vowels," *The Journal of the Acoustical Society of America*, vol. 97, no. 5, pp. 3099–3111, 1995.
- [17] S. Revoile, J. M. Pickett, L. D. Holden-Pitt, D. Talkin, and F. D. Brandt, "Burst and Transition Cues to Voicing Perception for Spoken Initial Stops by Impaired- and Normal-Hearing Listeners," *Journal of Speech, Language, and Hearing Research*, vol. 30, no. 1, pp. 3–12, 1987.
- [18] P. J. Jackson, "Acoustic cues of voiced and voiceless plosives for determining place of articulation," in *Workshop on Consistent and Reliable Acoustic Cues for sound analysis*, 2001.
- [19] A. Jongman, R. Wayland, and S. Wong, "Acoustic characteristics of English fricatives," *The Journal of the Acoustical Society of America*, vol. 108, no. 3, pp. 1252–1263, 2000.
- [20] K. Maniwa, A. Jongman, and T. Wade, "Acoustic characteristics of clearly spoken English fricatives," *The Journal of the Acoustical Society of America*, vol. 125, no. 6, 2009.
- [21] A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, and R. Pang, "Conformer: Convolution-augmented Transformer for Speech Recognition," in *Interspeech*, 2020.
- [22] Y. Jadoul, B. Thompson, and B. de Boer, "Introducing Parselmouth: A Python interface to Praat," *Journal of Phonetics*, vol. 71, pp. 1–15, 2018.
- [23] D. Can, V. R. Martinez, P. Papadopoulos, and S. S. Narayanan, "Pykaldi: A Python Wrapper for Kaldi," in *ICASSP*, 2018.
- [24] R. Sennrich, B. Haddow, and A. Birch, "Neural Machine Translation of Rare Words with Subword Units," in *ACL*, 2016.
- [25] T. Kudo and J. Richardson, "SentencePiece: A simple and language independent subword tokenizer and detokenizer for Neural Text Processing," in *EMNLP*, 2018.
- [26] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is All you Need," in *NeurIPS*, 2017.
- [27] C. Wang, Y. Tang, X. Ma, A. Wu, D. Okhonko, and J. Pino, "Fairseq S2T: Fast Speech-to-Text Modeling with Fairseq," in *AACL-IJCNLP*, 2020.
- [28] R. Ardila, M. Branson, K. Davis, M. Kohler, J. Meyer, M. Henretty, R. Morais, L. Saunders, F. Tyers, and G. Weber, "Common Voice: A Massively-Multilingual Speech Corpus," in *LREC*, 2020.
- [29] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: An ASR corpus based on public domain audio books," in *ICASSP*, 2015.
- [30] F. Hernandez, V. Nguyen, S. Ghannay, N. Tomashenko, and Y. Estève, "TED-LIUM 3: Twice as Much Data and Corpus Repartition for Experiments on Speaker Adaptation," in *Speech and Computer*, 2018.
- [31] C. Wang, M. Riviere, A. Lee, A. Wu, C. Talnikar, D. Haziza, M. Williamson, J. Pino, and E. Dupoux, "VoxPopuli: A Large-Scale Multilingual Speech Corpus for Representation Learning, Semi-Supervised Learning and Interpretation," in *ACL-IJCNLP*, 2021.
- [32] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the Inception Architecture for Computer Vision," in *CVPR*, 2016.
- [33] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, "Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks," in *ICML*, 2006.
- [34] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in *ICLR*, 2015.
- [35] D. S. Park, W. Chan *et al.*, "SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition," in *Interspeech*, 2019.
- [36] J. Garofolo, L. Lamel, W. Fisher, J. Fiscus, D. Pallett, and N. Dahlgren, "Darpa timit acoustic-phonetic continuous speech corpus cd-rom TIMIT," 1993-02-01 1993.
- [37] G. Attanasio, B. Savoldi, D. Fucci, and D. Hovy, "Twists, humps, and pebbles: Multilingual speech recognition models exhibit gender performance gaps," in *EMNLP*, 2024.
- [38] E. Chodroff and C. Wilson, "Burst spectrum as a cue for the stop voicing contrast in American English," *The Journal of the Acoustical Society of America*, vol. 136, no. 5, pp. 2762–2772, 2014.